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POWERING THE OCEAN'S FUTURE: A TOPOLOGY-DRIVEN HYDRODYNAMIC STUDY FOR FLOATING NUCLEAR POWER PLATFORMS

INTRODUCTION

Nuclear energy offshore could help Singapore meet its growing energy needs while working towards a more sustainable, low-carbon future. By moving nuclear facilities beyond the constraints of limited land, offshore platforms offer a new way to provide reliable, large-scale clean energy while preserving valuable space onshore. This poster provides an engineering solution to floating barges topologies to support offshore nuclear power operations.

PROBLEM STATEMENT

The number of nuclear powerplant designs in the world are limited. Akademik Lomonosov remains the only operational floating nuclear power plant in the world. There is a lack of safe innovative engineering designs specifically for offshore nuclear power plants.



PROOF OF CONCEPT

Introducing the HEXAGON© Offshore Nuclear Floating Platform

The HEXAGON© enables safe, state-of-the-art offshore nuclear energy operations, delivering the stability needed for reliable power generation without sacrificing valuable deck space.

Designed for deployment without boundaries, the platform can operate nearshore or in the open ocean. Its straight-edged hull enables direct integration with existing quays, bringing large-scale, reliable energy closer to where it is needed.

Combining state-of-the-art nuclear technology with integrated energy storage, the HEXAGON© goes beyond power generation. Its innovative design allows battery-powered vessels to dock alongside and recharge directly, transforming the platform into both a floating power station and a versatile offshore energy hub.

METHODOLOGY AND RESULTS

1) Modelling of barge shapes in SOLIDWORKS .

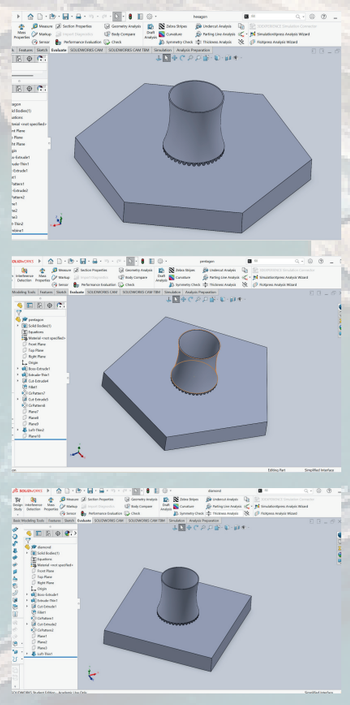
2) Performing of hydrodynamic diffraction analysis on Ansys .

3) Comparing the RAO outputs between different barge geometries (Hexagon, Pentagon , Square).

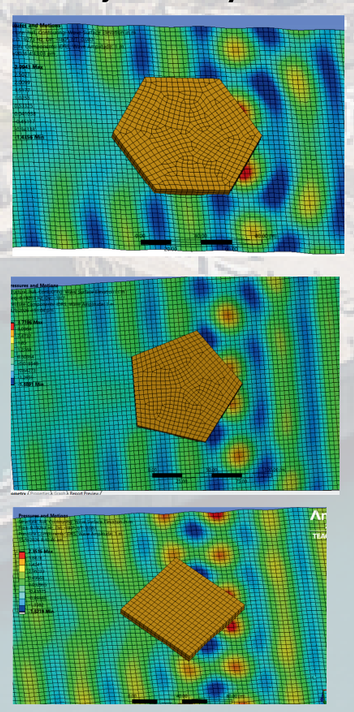
4) Based on the simulation results on Ansys, selecting the most suitable barge shape for offshore nuclear power operations.

$$\text{Hull geometry} \rightarrow \begin{cases} A(\omega) & \text{added mass} \\ B(\omega) & \text{radiation damping} \\ C & \text{hydrostatic restoring} \\ F(\omega) & \text{wave excitation} \end{cases} \rightarrow \xi(\omega) \rightarrow RAO$$

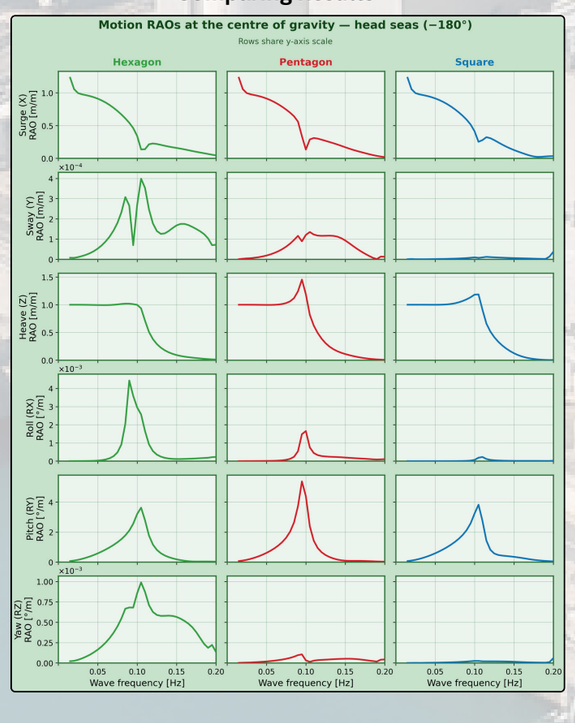
Basic design in SOLIDWORKS



Analysis in Ansys



Comparing Results



The figures above show the original engineering and computational processes behind selecting a suitable barge shape for an offshore nuclear power plant. The graphs on the right shows that the HEXAGON shaped barge has the best motion response compared to the other 2 whilst the PENTAGON shape to have the worst motion response.

Impacts

Engineering Impact
Improved 6dof motion response
Improved operational, mechanical and structural safety
Reduces production complexity and costs associated
Improves operational flexibility

Economic Impact
Supports marine decarbonization efforts
Supplements energy demands of countries / regions

Future Work

Offshore Engineering
Further hydrodynamic diffraction analysis
Hydrodynamic response studies incorporating Quadratic Tranfer Functions (QTF) and irregular wave spectrums (ie; JONSWAP, Pierson-Moskowitz Spectrums
Mooring analysis
Weight control study
Model studies

Nuclear Engineering
Plant design study
Nuclear safety

SIT DNA

Thinking thinkers
Think, experiment and solve creatively

Catalyst for transformation
Innovate, inspire and drive change

Able to learn, unlearn and relearn
Adapt, learn and continuously improve

Grounded in the community
Connect, contribute and create impact